MID-TERM/Minor EXAMINATION

(March, 2022)

Subject Code: BIT 202	**Subject: OPERATING SYSTEMS**

Time : 1 ½ Hours **Maximum Marks : 30**

Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q 1 **(2.5*4)**

(a) "Multi-tasking is an extension of multi-programming ." Comment whether this statement is true or false and give valid explanation for your choice of answer .

(b) What are the various states that a process can be in? Show with help of a process state diagram . Also , explain briefly about different types of schedulers .

(c) What are the different types of fragmentation that can occur in contiguous memory allocation ? What leads to the occurrence of the problem of fragmentation ? What kind of fragmentation does paging suffer from ? Explain briefly.

(d) Write short note on Real time operating system . Differentiate between hard real time OS and soft real time OS.

Q 2 **(5,5)**

(a) For the following set of processes, find the average waiting time & average turn around time using GANTT Chart such that the shortest job is executed first completely .

Process	Arrival Time (in secs)	Burst Time(in secs)
P1	0	4
P2	1	2
P3	2	5
P4	3	4

(b) Explain the concept of preemption in the context of CPU scheduling . Show how the introduction of preemption will affect the average waiting time and average turn around time in the above question i.e. 2(a).

Q 3 **(5,5)**

(a) Four jobs A,B,C and D arrive at a computer centre at the same time in this sequence. They have estimated running times of 9,5,3 and 4 secs. Their priorities are 3,4,2 and 1 respectively, with 1 being the highest priority. Find average waiting and average turnaround time using priority CPU scheduling when there is no preemption. Will the result vary for the given question if preemption was introduced ?

(b) Describe the LRU page replacement algorithm, assuming there are 3 frames and the page reference string is :
7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1 . Find the number of page faults that occur.

Q 4 **(5,5)**

(a) For the following set of processes, find the average waiting time & average turn around time using GANTT Chart. Assume that system is multiprocessing and each job gets its fair share of the CPU(time quantum= 2 secs)

Process	Arrival Time (in secs)	Burst Time(in secs)
P1	0.	5
P2	2.	2
P3	4	5
P4	6	4

(b) Consider the following page reference string 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6
Find out the number of page faults if there are 4 frames, using the optimal replacement algorithm . Why is the optimal page replacement considered better than the other page replacement algorithms?